

Codeforces Round 898 (Div. 4)

A. Short Sort

1 second, 256 megabytes

There are three cards with letters a , b , c placed in a row in some order. You can do the following operation **at most once**:

- Pick two cards, and swap them.

Is it possible that the row becomes abc after the operation? Output "YES" if it is possible, and "NO" otherwise.

Input

The first line contains a single integer t ($1 \leq t \leq 6$) — the number of test cases.

The only line of each test case contains a single string consisting of each of the three characters a , b , and c exactly once, representing the cards.

Output

For each test case, output "YES" if you can make the row abc with at most one operation, or "NO" otherwise.

You can output the answer in any case (for example, the strings "yEs", "yes", "Yes" and "YES" will be recognized as a positive answer).

input

```
6
abc
acb
bac
bca
cab
cba
```

output

```
YES
YES
YES
NO
NO
YES
```

In the first test case, we don't need to do any operations, since the row is already abc .

In the second test case, we can swap c and b : $acb \rightarrow abc$.

In the third test case, we can swap b and a : $bac \rightarrow abc$.

In the fourth test case, it is impossible to make abc using **at most one** operation.

B. Good Kid

1 second, 256 megabytes

Slavic is preparing a present for a friend's birthday. He has an array a of n digits and the present will be the product of all these digits. Because Slavic is a good kid who wants to make the biggest product possible, he wants to add 1 to exactly one of his digits.

What is the maximum product Slavic can make?

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \leq n \leq 9$) — the number of digits.

The second line of each test case contains n space-separated integers a_i ($0 \leq a_i \leq 9$) — the digits in the array.

Output

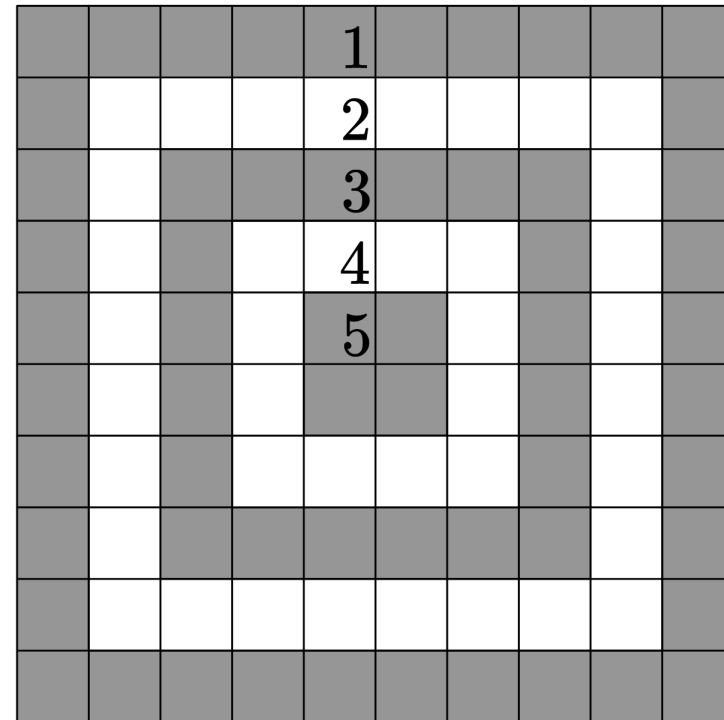
For each test case, output a single integer — the maximum product Slavic can make, by adding 1 to exactly one of his digits.

input
4
4
2 2 1 2
3
0 1 2
5
4 3 2 3 4
9
9 9 9 9 9 9 9 9 9
output
16
2
432
430467210

C. Target Practice

1 second, 256 megabytes

A 10×10 target is made out of five "rings" as shown. Each ring has a different point value: the outermost ring — 1 point, the next ring — 2 points, ..., the center ring — 5 points.



Vlad fired several arrows at the target. Help him determine how many points he got.

Input

The input consists of multiple test cases. The first line of the input contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

Each test case consists of 10 lines, each containing 10 characters. Each character in the grid is either X (representing an arrow) or . (representing no arrow).

Output

For each test case, output a single integer — the total number of points of the arrows.

input
8
6 3
WBWWWB
7 3
WWBWBWW
5 4
BWBWB
5 5
BBBBB
8 2
BWBWB
10 2
WBBWBWBWBW
4 1
BBBB
3 2
WWW
output
2
1
2
1
4
3
4
0

In the first test case you can perform the following operations:

WBWWWB → WWWWB → WWWWW

In the second test case you can perform the following operations:

WWBWBW → WWWWW

In the third test case you can perform the following operations:

BWBWB → BWWWW → WWWWW

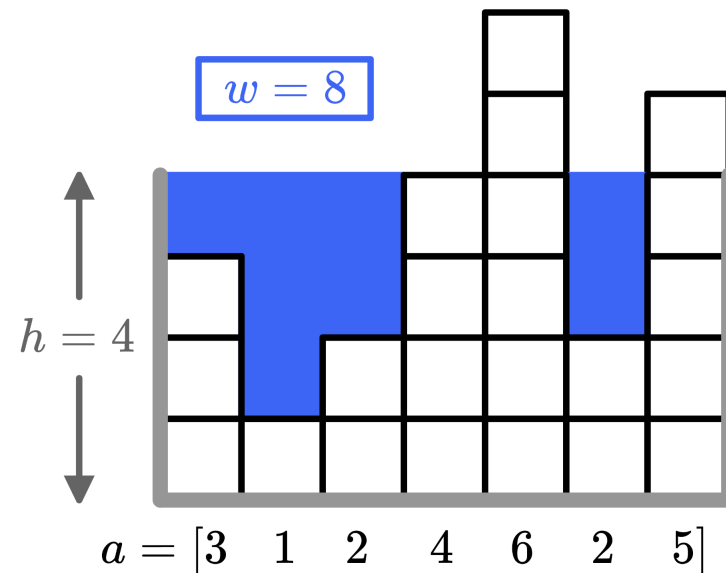
E. Building an Aquarium

2 seconds, 256 megabytes

You love fish, that's why you have decided to build an aquarium. You have a piece of coral made of n columns, the i -th of which is a_i units tall. Afterwards, you will build a tank around the coral as follows:

- Pick an integer $h \geq 1$ — the *height* of the tank. Build walls of height h on either side of the tank.
- Then, fill the tank up with water so that the height of each column is h , unless the coral is taller than h ; then no water should be added to this column.

For example, with $a = [3, 1, 2, 4, 6, 2, 5]$ and a height of $h = 4$, you will end up using a total of $w = 8$ units of water, as shown.



You can use at most x units of water to fill up the tank, but you want to build the biggest tank possible. What is the largest value of h you can select?

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains two positive integers n and x ($1 \leq n \leq 2 \cdot 10^5$; $1 \leq x \leq 10^9$) — the number of columns of the coral and the maximum amount of water you can use.

The second line of each test case contains n space-separated integers a_i ($1 \leq a_i \leq 10^9$) — the heights of the coral.

The sum of n over all test cases doesn't exceed $2 \cdot 10^5$.

Output

For each test case, output a single positive integer h ($h \geq 1$) — the maximum height the tank can have, so you need at most x units of water to fill up the tank.

We have a proof that under these constraints, such a value of h always exists.

input
5 7 9 3 1 2 4 6 2 5 3 10 1 1 1 4 1 1 4 3 4 6 1984 2 6 5 9 1 8 1 1000000000 1
output
4 4 2 335 1000000001

The first test case is pictured in the statement. With $h = 4$ we need 8 units of water, but if h is increased to 5 we need 13 units of water, which is more than $x = 9$. So $h = 4$ is optimal.

In the second test case, we can pick $h = 4$ and add 3 units to each column, using a total of 9 units of water. It can be shown that this is optimal.

In the third test case, we can pick $h = 2$ and use all of our water, so it is optimal.

F. Money Trees

2 seconds, 256 megabytes

Luca is in front of a row of n trees. The i -th tree has a_i fruit and height h_i .

He wants to choose a contiguous subarray of the array $[h_l, h_{l+1}, \dots, h_r]$ such that for each i ($l \leq i < r$), h_i is **divisible**[†] by h_{i+1} . He will collect all the fruit from each of the trees in the subarray (that is, he will collect $a_l + a_{l+1} + \dots + a_r$ fruits). However, if he collects more than k fruits in total, he will get caught.

What is the maximum length of a subarray Luca can choose so he doesn't get caught?

[†] x is *divisible* by y if the ratio $\frac{x}{y}$ is an integer.

Input

The first line contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

The first of each test case line contains two space-separated integers n and k ($1 \leq n \leq 2 \cdot 10^5$; $1 \leq k \leq 10^9$) — the number of trees and the maximum amount of fruits Luca can collect without getting caught.

The second line of each test case contains n space-separated integers a_i ($1 \leq a_i \leq 10^4$) — the number of fruits in the i -th tree.

The third line of each test case contains n space-separated integers h_i ($1 \leq h_i \leq 10^9$) — the height of the i -th tree.

The sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case output a single integer, the length of the maximum length contiguous subarray satisfying the conditions, or 0 if there is no such subarray.

input
5 5 12 3 2 4 1 8 4 4 2 4 1 4 8 5 4 1 2 6 2 3 1 3 12 7 9 10 2 2 4 1 10 11 1 7 10 2 6 3 1 5 10 6 72 24 24 12 4 4 2
output
3 2 1 0 3

In the first test case, Luca can select the subarray with $l = 1$ and $r = 3$.

In the second test case, Luca can select the subarray with $l = 3$ and $r = 4$.

In the third test case, Luca can select the subarray with $l = 2$ and $r = 2$.

G. ABBC or BACB

1 second, 256 megabytes

You are given a string s made up of characters A and B. Initially you have no coins. You can perform two types of operations:

- Pick a substring[†] AB, change it to BC, and get a coin.
- Pick a substring[†] BA, change it to CB, and get a coin.

What is the most number of coins you can obtain?

[†] A *substring* of length 2 is a sequence of two adjacent characters of a string.

Input

The input consists of multiple test cases. The first line of the input contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

The only line of each test case contains the string s ($1 \leq |s| \leq 2 \cdot 10^5$). All characters of s are either A or B.

The sum of the lengths of s over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the maximum number of coins you can obtain.

input
8 ABBA ABA BAABA ABB AAAAAAB BABA B AAA

output
2
1
3
1
6
2
0
0

In the first test case you can perform the following operations to get 2 coins:

ABBA → BCBA → BCCB

In the second test case you can perform the following operation to get 1 coin:

ABA → BCA

In the third test case you can perform the following operations to get 3 coins:

BAABA → CBABA → CBACB → CCBCB

H. Mad City

4 seconds, 256 megabytes

Marcel and Valeriu are in the mad city, which is represented by n buildings with n two-way roads between them.

Marcel and Valeriu start at buildings a and b respectively. Marcel wants to catch Valeriu, in other words, be in the same building as him or meet on the same road.

During each move, they choose to go to an adjacent building of their current one or stay in the same building. Because Valeriu knows Marcel so well, Valeriu can predict where Marcel will go in the next move. Valeriu can use this information to make his move. They start and end the move at the same time.

It is guaranteed that any pair of buildings is connected by some path and there is at most one road between any pair of buildings.

Assuming both players play optimally, answer if Valeriu has a strategy to indefinitely escape Marcel.

Input

The first line contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

The first line of each test case contains three space-separated integers n , a , b ($3 \leq n \leq 2 \cdot 10^5$; $1 \leq a, b \leq n$) — the number of buildings (which equals the number of roads) and the starting buildings of Marcel and Valeriu.

The following n lines each contain two integers u_i , v_i ($1 \leq u_i, v_i \leq n$, $u_i \neq v_i$) — there is a road between buildings u_i and v_i . There is at most one road between any unordered pair of buildings.

The sum of n over all test cases does not exceed $2 \cdot 10^5$.

The roads are given that it is possible to get from any building to any other building going along the roads.

Output

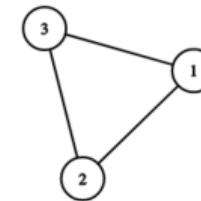
For each test case output "YES" if Valeriu can escape Marcel forever and "NO" otherwise.

You can output the answer in any case (for example, the strings "yEs", "yes", "Yes" and "YES" will be recognized as a positive answer).

input
6
3 2 1
2 1
3 2
1 3
4 1 4
1 4
1 2
1 3
2 3
4 1 2
1 2
2 3
2 4
3 4
7 1 1
4 1
2 1
5 3
4 6
4 2
7 5
3 4
8 5 3
8 3
5 1
2 6
6 8
1 2
4 8
5 7
6 7
10 6 1
1 2
4 3
5 8
7 8
10 4
1 9
2 4
8 1
6 2
3 1

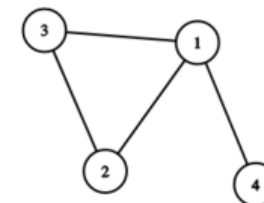
output
YES
NO
YES
NO
NO
YES

In the first test case the graph looks as follows:



Marcel starts at building 2, while Valeriu starts at building 1. Valeriu knows which way Marcel will move around the triangle, and he can simply always move in the same way to avoid Marcel forever.

In the second test case the graph looks as follows:



Marcel starts at building 1, while Valeriu starts at building 4. Marcel can go to building 4 on his first move and win, since Valeriu must either go to building 1 (then he meets Marcel on the road from 1 to 4) or stay at building 4 (then he meets Marcel at building 4). So there is no strategy for Valeriu to win.

